

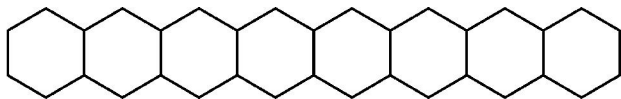


Series : W4YXZ

SET ~ 5

रोल नं.

Roll No.



प्रश्न-पत्र कोड
Q.P. Code **56(B)**

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

नोट / NOTE



- (I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 23 हैं।
Please check that this question paper contains 23 printed pages.
- (II) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 33 प्रश्न हैं।
Please check that this question paper contains 33 questions.
- (III) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- (IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में यथा स्थान पर प्रश्न का क्रमांक अवश्य लिखें।
Please write down the serial number of the question in the answer-book at the given place before attempting it.
- (V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक परीक्षार्थी केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.
- ~



रसायन विज्ञान (सैद्धांतिक)
(केवल दृष्टिबाधित परीक्षार्थियों के लिए)



CHEMISTRY (Theory)

(FOR VISUALLY IMPAIRED CANDIDATES ONLY)

निर्धारित समय : 3 घण्टे

अधिकतम अंक : 70

Time allowed : 3 hours

Maximum Marks : 70

56(B)

738

1

P.T.O.



General Instructions :

Read the following instructions carefully and follow them :

- (i) This Question paper contains **33** questions. **All** questions are **compulsory**.
- (ii) This Question paper is divided into **five** sections – **Section A, B, C, D and E**.
- (iii) **Section A** – Question number **1 to 16** are multiple choice type questions. Each question carries **1** mark.
- (iv) **Section B** – Question number **17 to 21** are very short answer type questions. Each question carries **2** marks.
- (v) **Section C** – Question number **22 to 28** are short answer type questions. Each question carries **3** marks.
- (vi) **Section D** – Question number **29 and 30** are case-based questions. Each question carries **4** marks.
- (vii) **Section E** – Question number **31 to 33** are long answer type questions. Each question carries **5** marks.
- (viii) There is no overall choice given in the question paper. However, an internal choice has been provided in few questions in all the sections except **Section A**.
- (ix) Use of calculator is **not** allowed.

SECTION – A

1. Which of the following is a polysaccharide ?
- | | |
|---------------|-------------|
| (A) Maltose | (B) Glucose |
| (C) Cellulose | (D) Sucrose |



2. The Gabriel Phthalimide Synthesis is used for the preparation of
- (A) Secondary amines (B) Primary aromatic amines
(C) Tertiary amines (D) Primary aliphatic amines
3. The compound which undergoes dehydration most easily is
- (A) 2-Methylpropan -2- ol (B) Ethanol
(C) 2-Methylbutan -2- ol (D) Propan -1- ol
4. The conversion of an alkyl halide into an alcohol by aqueous NaOH is classified as
- (A) an addition reaction
(B) a substitution reaction
(C) a dehydrohalogenation reaction
(D) a dehydration reaction
5. Which of the following transition metals does not show variable oxidation states ?
- (A) Cu (B) Sc
(C) Ti (D) Fe
6. An azeotropic mixture of two liquids has a boiling point higher than that of either of the two liquids when it
- (A) obeys Raoult's law.
(B) shows positive deviation from Raoult's law.
(C) shows negative deviation from Raoult's law.
(D) obeys Henry's law.



7. Kohlrausch gave the following relation for strong electrolytes at low concentration :

$$\Lambda_m = \Lambda_m^0 - A\sqrt{C}$$

Which of the following equality holds true ?

- (A) $\Lambda_m = \Lambda_m^0$ as $C \rightarrow \sqrt{A}$ (B) $\Lambda_m = \Lambda_m^0$ as $C \rightarrow 0$
(C) $\Lambda_m = \Lambda_m^0$ as $C \rightarrow \infty$ (D) $\Lambda_m = \Lambda_m^0$ as $C \rightarrow 1$
8. The rate of reaction $X + Y \rightarrow \text{products}$, is given by the equation
Rate = $k[X][Y]$.
If Y is taken in large excess, the order of the reaction would be
(A) 0 (B) 1
(C) 2 (D) $\frac{1}{2}$
9. The synthesis of alkyl fluoride is best accomplished by
(A) Sandmeyer reaction (B) Finkelstein reaction
(C) Wurtz reaction (D) Swarts reaction
10. Benzene diazonium chloride on hydrolysis gives
(A) Chlorobenzene (B) Phenol
(C) Anisole (D) Aniline
11. Solubility of gases in liquid decreases with increase in
(A) pressure (B) volume
(C) number of solute particles (D) temperature



12. $\Delta_r G^0$ and E_{cell}° for a spontaneous reaction will respectively be
- (A) positive and positive (B) positive and negative
(C) negative and positive (D) negative and negative

For question number **13** to **16**, two statements are given, one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below :

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.
13. **Assertion (A) :** Zinc is not regarded as a transition element.
Reason (R) : Zinc has completely filled 3d orbitals in its ground state as well as in its oxidised state.
14. **Assertion (A) :** In a first order reaction, if the concentration of the reactant is doubled, its half-life is also doubled.
Reason (R) : The half-life of a reaction does not depend upon the initial concentration of the reactant in a first order reaction.



15. **Assertion (A)** : $(\text{CH}_3)_3\text{C}-\text{O}-\text{CH}_3$ gives $(\text{CH}_3)_3\text{C}-\text{I}$ and CH_3OH on treatment with HI .

Reason (R) : The reaction occurs by $\text{S}_{\text{N}}1$ mechanism.

16. **Assertion (A)** : Nucleophilic substitution reaction of chlorobenzene is easier than that of chloroethane.

Reason (R) : $\text{C}-\text{Cl}$ bond in chlorobenzene has partial double bond character due to resonance.

SECTION – B

17. (a) Why does the cell potential of mercury cell remains constant throughout its life ?

(b) Calculate the degree of dissociation (α) of CH_3COOH if

Λ_{m} and Λ_{m}^0 of CH_3COOH are $50 \text{ S cm}^2 \text{ mol}^{-1}$ and $400 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. **2 × 1 = 2**

18. Using the E^0 values of P and Q, predict which one is better for coating the surface of iron [$E^0_{(\text{Fe}^{2+}/\text{Fe})} = -0.44 \text{ V}$] to prevent corrosion and why ?

$E^0_{(\text{P}^{2+}/\text{P})} = -2.37 \text{ V}$, $E^0_{(\text{Q}^{2+}/\text{Q})} = -0.14 \text{ V}$ **2**

19. (a) A reaction is second order in 'A' and first order in 'B'. How is the rate affected when the concentrations of both 'A' and 'B' are doubled ?

(b) Write the unit of 'k' for zero order reaction. **2 × 1 = 2**



20. (a) How do you convert the following ?
(i) Ethanenitrile to Ethanamine
(ii) Benzenediazonium chloride to benzonitrile $2 \times 1 = 2$

OR

- (b) (i) Write a simple chemical test to distinguish between ethanamine and dimethylamine.
(ii) What happens when CH_3CONH_2 is heated with Br_2 and an aqueous solution of NaOH ? $2 \times 1 = 2$
21. What is the effect of denaturation on the structures of protein ? Give one example each of fibrous protein and globular protein. **2**

SECTION – C

22. What happens when
(a) Ethanal is treated with CH_3MgBr followed by hydrolysis ?
(b) Phenol is treated with Zinc dust ?
(c) Anisole is treated with HI ? $3 \times 1 = 3$
23. The rate constant of a first order reaction increases from 0.04 s^{-1} to 0.08 s^{-1} when the temperature increases from 27°C to 37°C . Calculate the energy of activation (E_a).
[Given : $2.303R = 19.15 \text{ JK}^{-1}\text{mol}^{-1}$]
[$\log 2 = 0.3010$, $\log 3 = 0.4771$, $\log 4 = 0.6021$] **3**
24. Give reactions for the following :
(i) Aniline does not undergo Friedel – Crafts reaction.
(ii) $(\text{CH}_3)_2\text{NH}$ is more basic than $(\text{CH}_3)_3\text{N}$ in an aqueous solution.
(iii) Ethyl amine is soluble in water whereas aniline is insoluble. $3 \times 1 = 3$



25. (a) When a co-ordination compound $\text{CoCl}_3 \cdot 6\text{NH}_3$ is mixed with excess of AgNO_3 solution, 3 moles of AgCl are precipitated per mole of the compound. Write
- (i) Structural formula of the complex
 - (ii) IUPAC name of the complex
 - (iii) Hybridization of the complex using valence bond theory [Atomic number: Co = 27] $3 \times 1 = 3$

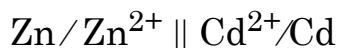
OR

- (b) (i) Write IUPAC name of $[\text{Pt}(\text{NH}_3)_2 \text{Cl}(\text{ONO})]$.
- (ii) Why $[\text{Co}(\text{en})_3]^{3+}$ is more stable complex than $[\text{Co}(\text{NH}_3)_6]^{3+}$?
- (iii) Predict the hybridization of $[\text{Ni}(\text{CO})_4]$ on the basis of valence bond theory. [Atomic number : Ni = 28] $3 \times 1 = 3$

26. Explain the following reactions :

- (a) Rosenmund's Reduction
- (b) Cannizzaro's reaction
- (c) Hell-Volhard Zelinsky reaction $3 \times 1 = 3$

27. Calculate $\Delta_r G^\circ$ and $\log K_C$ for the following cell at 25°C :



Given that : $E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76 \text{ V}$, $E_{\text{Cd}^{2+}/\text{Cd}}^0 = 0.40 \text{ V}$

$$1 \text{ F} = 96500 \text{ C mol}^{-1}. \quad \quad \quad \mathbf{3}$$

28. Define Racemization. Out of $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ reactions, which is accompanied by racemization ?

Give reason in support of your answer. $\mathbf{3}$



SECTION – D

The following questions are case based questions. Each question has an internal choice and carries 4 (2 + 1 + 1) marks each. Read the passage carefully and answer the questions that follow :

29. Carbohydrates are optically active polyhydroxy aldehydes or ketones or molecules which provide such units on hydrolysis. They are broadly classified into three groups – monosaccharides, oligosaccharides and polysaccharides. Monosaccharides are held together by glycosidic linkage to form disaccharides like sucrose, maltose.

Another biomolecule : proteins are polymers of α -amino acids which are linked by peptide bonds. On the basis of number of amino group and carboxyl group, amino acids are classified as acidic, basic or neutral amino acids. Amino acids are amphoteric in nature.

- (a) What is difference between glycosidic linkage and peptide linkage ? The pentacetate of glucose does not react with Hydroxyl amine. What does it indicate ? 2
- (b) Define Oligosaccharides with an example. 1

OR

- (b) Why amino acids are amphoteric in nature ? 1
- (c) Define Acidic amino acids. 1



30. Werner proposed the concept of a primary valence and a secondary valence for a metal ion. The primary valences are normally ionisable and are satisfied by negative ions. The secondary valences are non-ionisable. These are satisfied by neutral molecules or negative ions. The secondary valency is equal to the co-ordination number and is normally fixed for a metal. The Valence Bond Theory (VBT) explains the formation, magnetic behaviour and geometrical shapes of co-ordination compounds whereas the Crystal Field Theory (CFT) is based on the effect of different crystal fields on the degeneracy of d-orbitals energies of the central metal atom / ion.

- (a) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ is paramagnetic while $[\text{Ni}(\text{CN})_4]^{2-}$ is diamagnetic. Explain why ? [Atomic number : Cr = 24, Ni = 28] 2
- (b) Write one difference between a primary valence and a secondary valence. 1
- (c) What is crystal field splitting energy ? 1

OR

- (c) On the basis of CFT, write the electronic configuration of d^4 orbitals when $\Delta_0 > P$. 1

SECTION – E

31. (A) (a) Account for the following :
- (i) Transition metals and their compounds show catalytic activities.
 - (ii) Mn^{3+} is a strong oxidising agent.
 - (iii) Cu^+ is not stable in aqueous solution.
- (b) Write the preparation of KMnO_4 from Pyrolusite ore (MnO_2). 3 + 2 = 5

OR



- (B) (i) Write the preparation of $\text{Na}_2\text{Cr}_2\text{O}_7$, from FeCr_2O_4 .
(ii) What is Lanthanoid contraction ? Write its two consequences.
(iii) Name two elements of 3d series which show anomalous electronic configuration. **2 + 2 + 1 = 5**

32. (A) (a) 'A' and 'B' are two functional isomers of compound $\text{C}_4\text{H}_8\text{O}$. On heating with NaOH and I_2 , isomer 'B' forms yellow precipitate of iodoform whereas isomer 'A' does not form any precipitate.

- (i) Identify 'A' and 'B'.
(ii) What happens when isomer 'A' is treated with Zn(Hg) in the presence of Conc. HCl ?
(iii) Write the reaction of isomer 'B' with NaOH and I_2 .
(b) Arrange the following in the increasing order of their property as indicated :
(i) Ethanol, Ethanoic acid, Ethanal (boiling point).
(ii) Ethanal, Methanal, acetone
(reactivity towards addition of HCN). **3 + 2 = 5**

OR

- (B) (a) Explain Aldol Condensation with an example. Why α (α) hydrogen of aldehydes and ketones are acidic in nature ?
(b) Give simple chemical test to distinguish between the following compounds :
(i) Benzoic acid and Benzaldehyde
(ii) Ethanal and Propanal **3 + 2 = 5**



33. (A) (a) Calculate the boiling point of solution when 6 g of MgSO_4 (Molar mass = 120 g mol^{-1}) was dissolved in 200 g of water, assuming the complete dissociation of MgSO_4 . (K_b for water = $0.52 \text{ K kg mol}^{-1}$).

(b) State Raoult's law for a solution containing volatile components. How Raoult's law is a special case of Henry's law ?

3 + 2 = 5

OR

(B) (a) The freezing point of a solution containing 5 g of benzoic acid (Molar mass = 122 g mol^{-1}) in 35 g of benzene is depressed by 2.94 K. Calculate the percentage association of benzoic acid if it forms a dimer in solution. (K_f for benzene = $4.9 \text{ K kg mol}^{-1}$)

(b) Define the following terms :

(i) Ideal Solution

(ii) Osmotic Pressure

3 + 2 = 5

